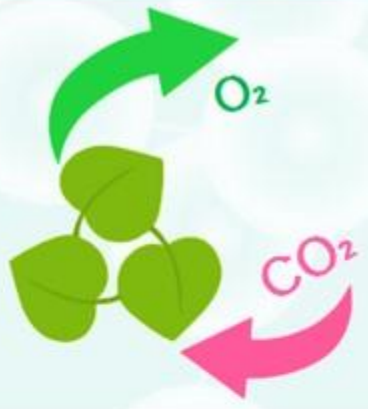




BIOLOGY



CONTROL AND COORDINATION

MODULE - 15



NERVOUS SYSTEM



- Complex network of nerve cells that carry messages
- To and fro from the Brain and the Spinal Cord
- Controls and coordinates the body functions
- Produces hormones
- Quick action in the form of reflexes

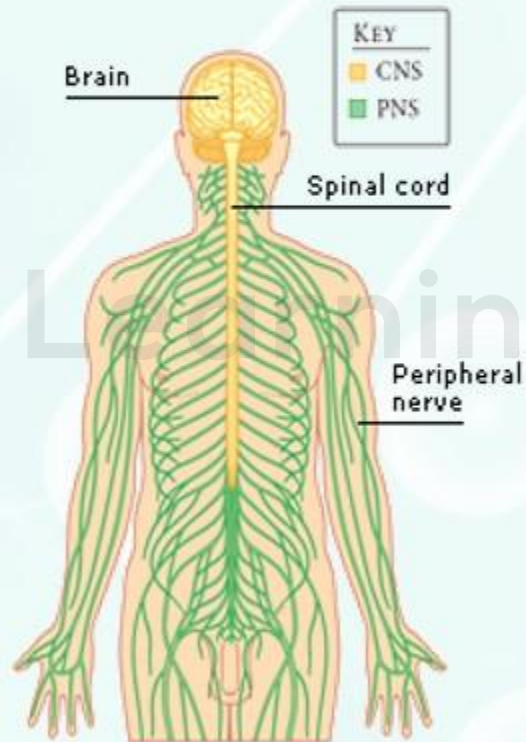
They are the first line of control and coordination in higher organisms



HUMAN NERVOUS SYSTEM

CENTRAL NERVOUS SYSTEM

- Consists of Brain and Spinal cord
- It receives the information from all parts of body for processing and integration

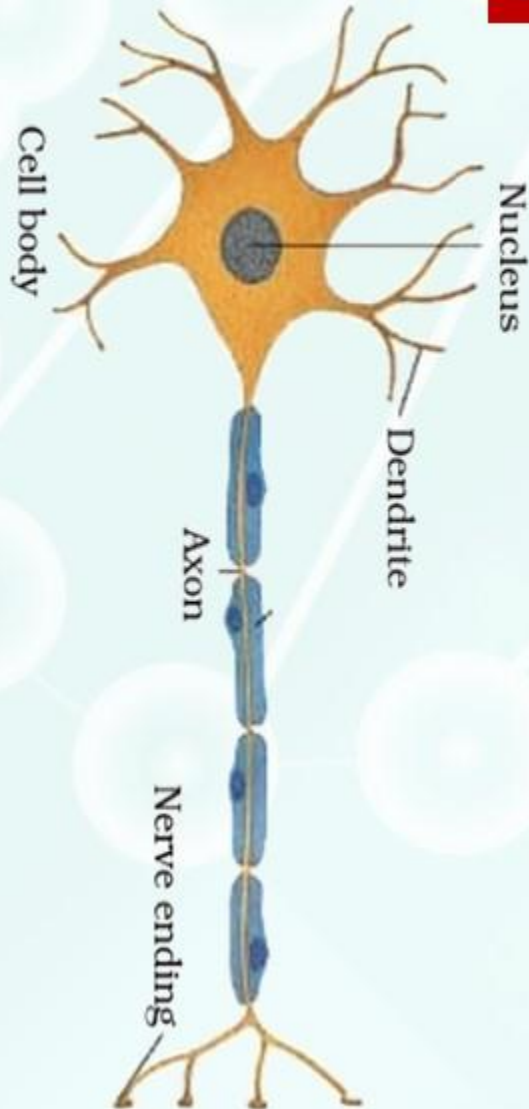


PERIPHERAL NERVOUS SYSTEM

- Lie outside the brain and spinal cord
- Connect CNS to limbs and organs
- Comprises of
 - Autonomic nervous system
 - Somatic nervous system



NEURON OR A NERVE CELL



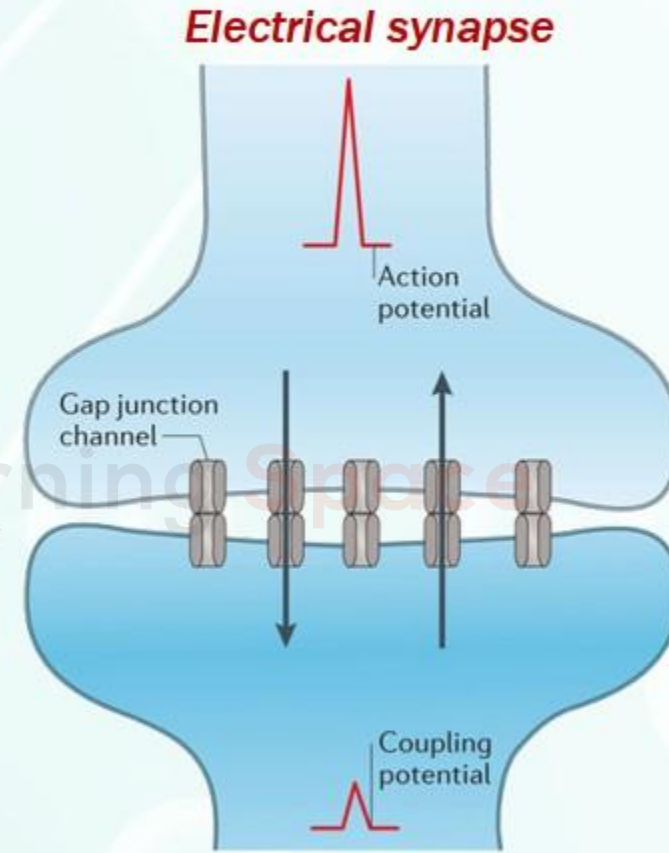
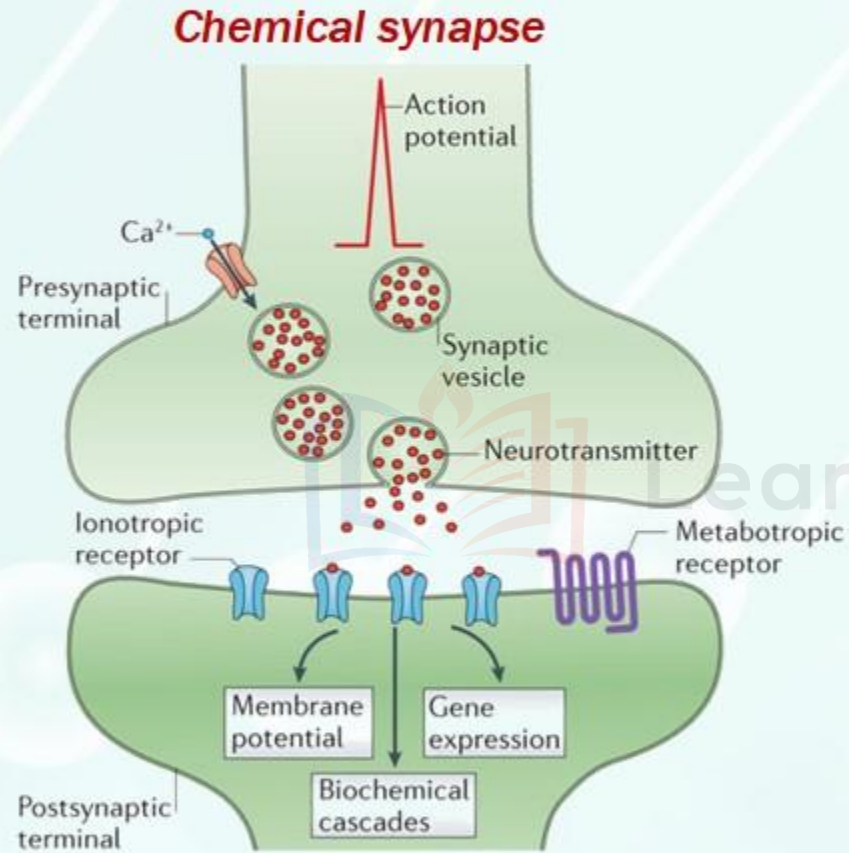
Neurons are the primary components of the central nervous system and the peripheral nervous system.

There are many types of specialized neurons.

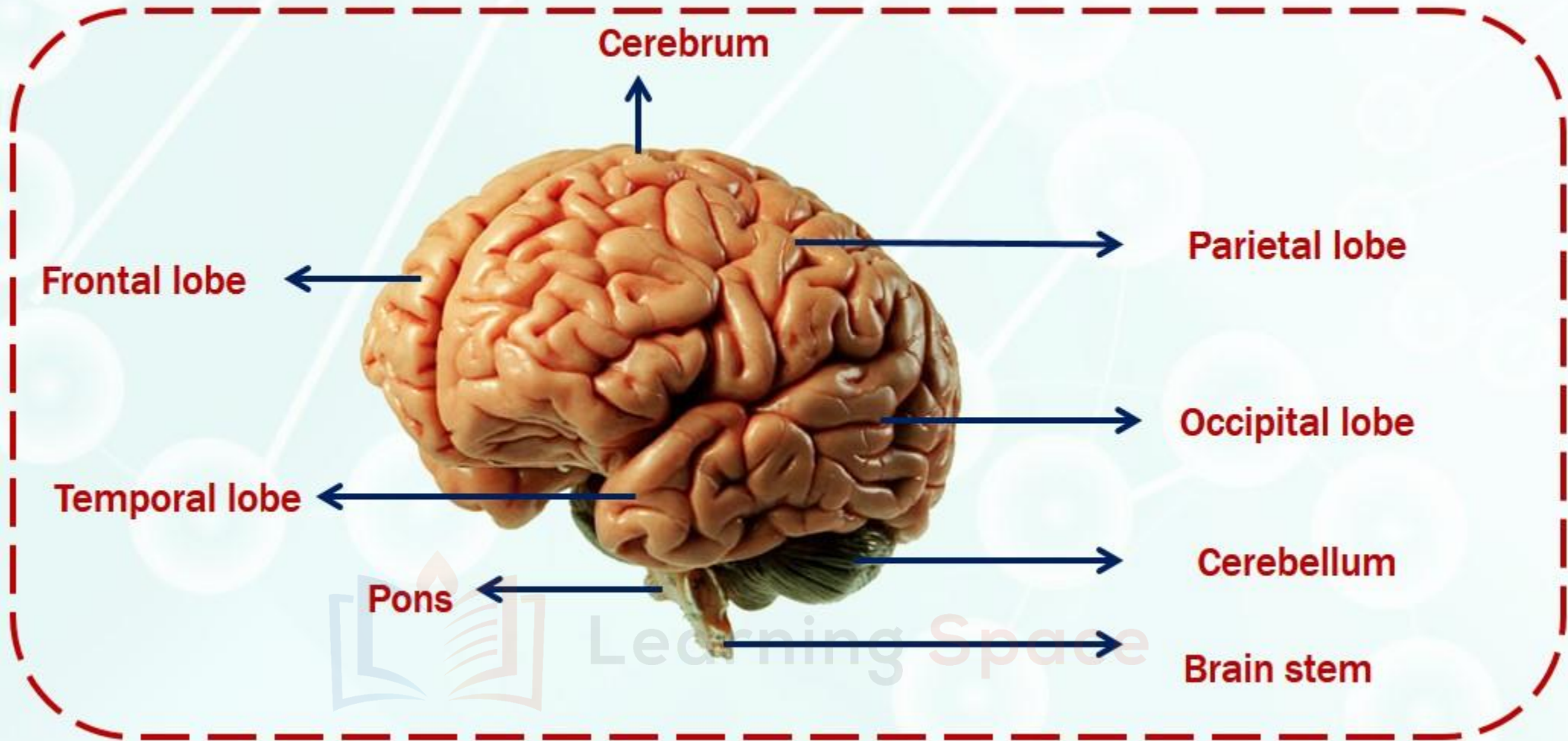
- 1) **SENSORY NEURONS** respond stimuli such as touch, sound, or light and all other stimuli affecting the cells of the sensory organs, and converts it into an electrical signal which is then sent to the spinal cord or brain.
- 2) **MOTOR NEURONS** receive signals from the brain and spinal cord to cause everything from muscle contractions and affect glandular outputs.
- 3) **INTERNEURONS** connect neurons to other neurons within the same region of the brain or spinal cord in neural networks.



HOW AN IMPULSE TRAVELS?



PARTS OF A HUMAN BRAIN



HUMAN BRAIN

CEREBRUM

- Touch
- Vision
- Hearing
- Speech
- Reasoning
- Emotions
- Learning
- Fine control movements

CEREBELLUM

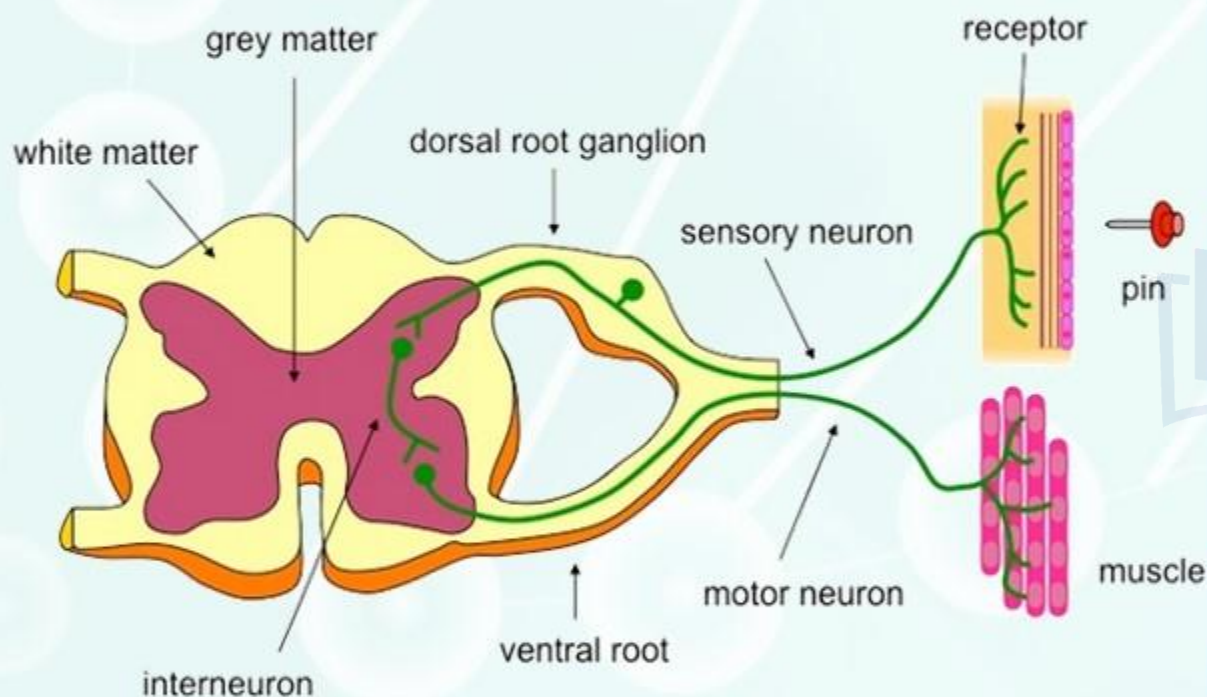
- Co-ordinate muscle movements
- Maintain posture
- Maintain balance

BRAIN STEM

- Relay center connecting to the cerebrum and cerebellum to the spinal cord
- Breathing
- Heart rate
- Body temperature
- Wake and sleep cycles
- Digestion
- Sneezing and swallowing
- Coughing and Vomiting



REFLEX ACTION



- When a receptor is stimulated, it sends a signal to the central nervous system, where the brain co-ordinates the response. But sometimes a very quick response is needed, one that does not need the involvement of the brain. This is a reflex action.
- Reflex actions are rapid and happen without us thinking.
- The actions involved are:
 - **receptor detects a stimulus - change in the environment**
 - **sensory neuron sends signal to relay neuron**
 - **motor neuron sends signal to effector**
 - **effector produces a response**



BRAIN DISORDERS

MENINGITIS



- Meningitis is an infection that affects and inflames the delicate membranes called meninges that cover the brain and spinal cord.
- Meningitis can be caused by bacteria, fungus and virus.
- The symptoms of this disease are sudden fever, severe headache, neck stiffness, joint pain or mental changes.
- Severe complications can occur such as paralysis, brain damage, deafness or death.
- Meningitis can be prevented by maintaining personal hygiene and getting a meningococcal vaccine.



BRAIN DISORDERS



Healthy Brain



Mild Alzheimer's Disease



Severe Alzheimer's Disease

ALZHEIMER'S DISEASE

- Alzheimer's disease is a gradual, irreversible disease that destroys memory and other mental functions.
- It is the most common cause of dementia.
- It is a neurodegenerative disease which is caused due to death of a neuron.
- The size of the brain starts to shrink.
- Patients develop memory lapses and develop problems such as thinking, reasoning, perception and communication.
- It may be prevented by certain lifestyle choices, physical activity and a healthy diet.



CHEMICAL COORDINATION

ENDOCRINE SYSTEM



Endocrine Glands



Hormones



Target organs



Desired body functions



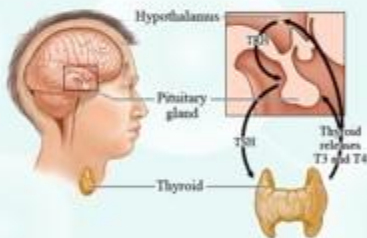
ENDOCRINE SYSTEM IN HUMANS



ADRENALINE

Adrenal glands

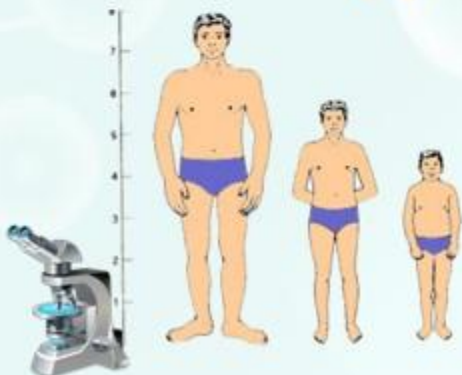
Faster heart beat, breathing rate increases, acts as an emergency hormone



THYROXINE

Thyroid gland

Regulate protein, fat, carbohydrate metabolism. Requires dietary iodine



GROWTH HORMONE

Pituitary gland

Growth and development of bones and overall body.

ENDOCRINE SYSTEM IN HUMANS



INSULIN

Pancreas

Helps in regulating blood sugar level



TESTOSTERONE

Leydig cells in the testes

Responsible for puberty changes in males



ESTROGEN

*Pituitary gland/
ovaries*

Responsible for puberty changes in females



COORDINATION IN PLANTS



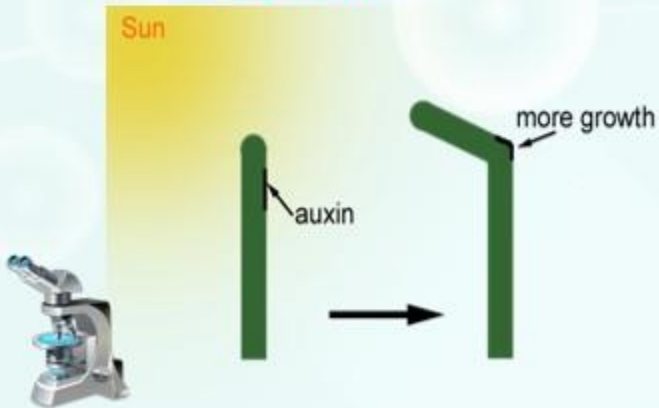
- Plants use a chemical system instead of a nervous system to respond to external stimulus.
- They respond to stimuli such as light, gravity, touch and water.
- Plants also show two different types of movements
 - Growth dependent
 - Growth independent



HORMONES IN PLANTS/ PHYTOHORMONES

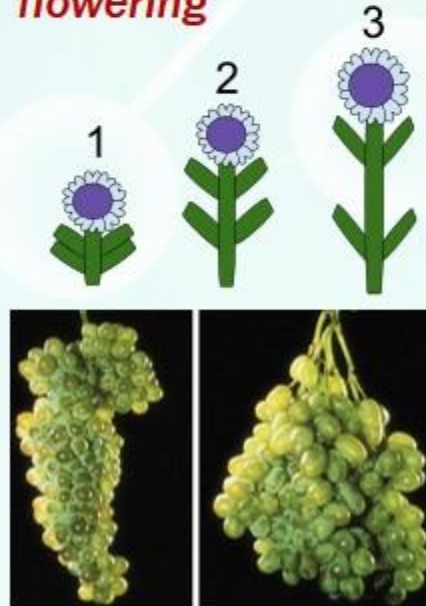
AUXINS

- It is synthesized at the tip of the stem.
- Help in plant growth by elongating the cell
- It diffuses towards the shady side of the root



GIBBERELLINS

- Concerned with the growth of the stem.
- Also promotes seed germination and flowering



CYTOKININS

- Present in areas of cell division
- Helps in opening up of the stomata



ABSCISIC ACID

- Inhibit the growth of many parts
- Closure of stomata
- Falling of leaves
- Bud dormancy

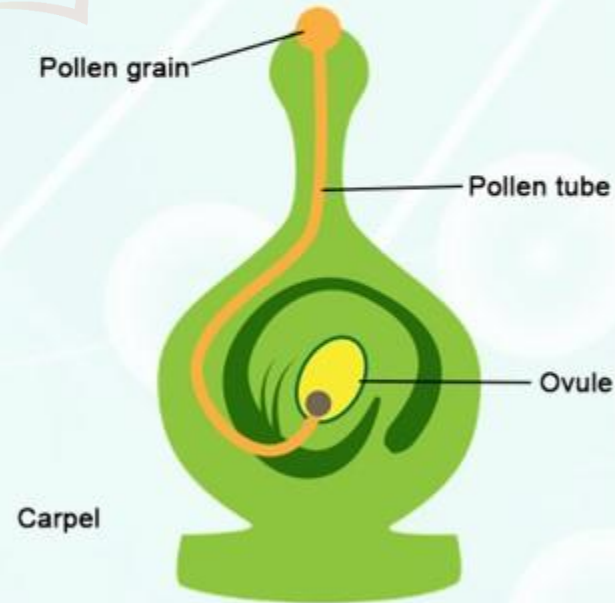


GROWTH DEPENDENT MOVEMENT

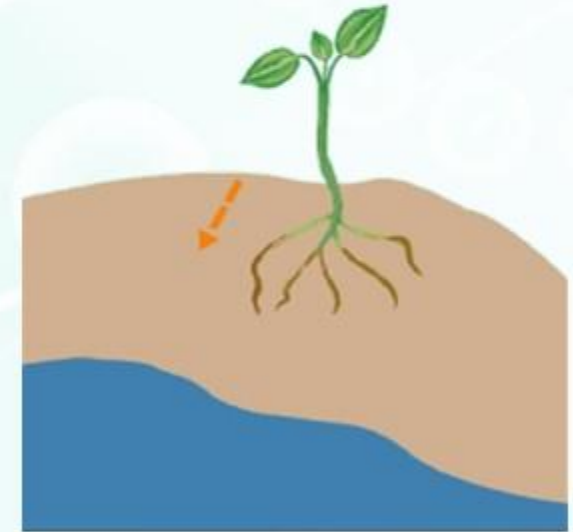
GEOTROPISM



CHEMOTROPISM

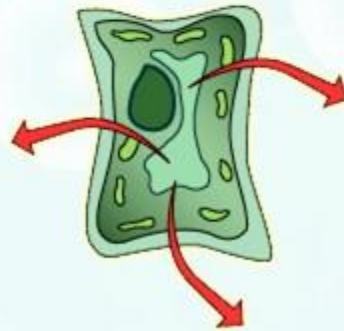
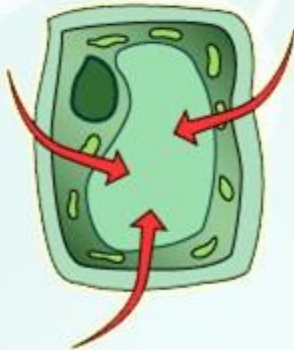


HYDROTROPISM



GROWTH INDEPENDENT MOVEMENT

Sensitive plant



- No growth is involved in such plants.
- The plant can move in response to an external stimulus.
- The plants use chemical and electrical means to convey the information from cell to cell.
- Plant cells change the amount of water in them, resulting in swelling and shrinking.



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